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Hybrid and transformer based artificial intelligence for diabetic retinopathy diagnosis a systematic review of methods challenges and clinical readiness

Frenisha Digaswala[1*] and Amit Ganatra[2]

*Correspondence:
Frenisha Digaswala
frenisha.digaswala22620@paruluniversity.ac.in
[1]Faculty of Engineering and Technology, Parul University, Vadodara, Gujarat, India
[2]Director R & D and Dean FET, Charutar Vidya Mandal (CVM) University, V V Nagar, Gujarat, India

Abstract

Diabetic retinopathy (DR) remains a leading cause of preventable blindness worldwide, placing a growing burden on healthcare systems. While artificial intelligence (AI) offers promising tools for early detection and scalable screening, its transition from research to real-world clinical use faces significant hurdles. This systematic review, guided by PRISMA principles, examines 27 recent studies (2023–2025) to map the evolving landscape of AI-driven DR diagnosis. We categorize approaches into five families: convolutional neural networks (CNNs), hybrid CNN–machine learning models, transformer-based architectures, clinical data–driven predictors, and multimodal fusion systems. Our analysis reveals that while transformers excel in severity grading and hybrid models demonstrate practical robustness, critical gaps persist—including poor early-DR sensitivity, limited generalizability across imaging devices, and a lack of clinical explainability. We argue that future efforts must prioritize lightweight, interpretable, and temporally aware AI systems that integrate multimodal patient data. By bridging technical innovation with clinical pragmatism, this review aims to accelerate the development of deployable AI solutions for global DR screening.

Keywords Diabetic retinopathy, Deep learning, Multimodal AI, Vision transformers, Explainable AI, Tele-ophthalmology, Clinical translation

1 Introduction

Diabetic retinopathy (DR) is a microvascular complication of diabetes that harms the retina and, if not treated, can cause permanent vision loss. With the number of people with diabetes around the world expected to rise from 463 million to more than 700 million by 2045, the need for quick and easy DR screening has never been more urgent [1]. Early detection is crucial in clinical practice—prompt intervention during non-proliferative phases can avert up to 90% of diabetes-related blindness.

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Ophthalmologists manually grade fundus images to diagnose DR in the traditional way. This method has problems with cost, inter-grader variability, and a lack of specialists in areas that don't have enough of them [2]. In response, AI-based screening systems have become revolutionary tools that can automate finding, grading, and even predicting how a disease will get worse.

In the last few years, there have been a lot of AI methods used in DR, such as basic CNNs [3–6], hybrid deep learning–ML frameworks [10–15], new transformer architectures [16, 17], clinical risk models [23–25], and multimodal fusion systems [18–22]. However, despite impressive performance metrics, the road to clinical integration is still blocked by high computational needs, a lack of interpretability, and a lack of generalization across different populations and imaging devices.

1.1 Motivation and scope

(i) While prior reviews have often organized DR-AI literature by dataset or clinical validation level, few have adopted a technique-centered lens to examine how architectural innovations directly impact diagnostic accuracy and clinical utility. This review fills that gap by categorizing recent work into five coherent families:
(ii) CNN-based models (spatial feature extraction)
(iii) Hybrid CNN + ML systems (deep features + traditional classifiers)
(iv) Transformer and attention-based methods (global context modeling)
(v) Clinical ML predictors (tabular health data)
(vi) Multimodal fusion approaches (imaging + clinical data integration)

1.2 Contributions

This paper provides:

- A structured analysis of 27 cutting-edge studies on DR detection and grading.
- A unified comparison of datasets, preprocessing techniques, model architectures, performance metrics, and computational requirements.
- Identification of persistent research challenges in generalization, explainability, resource-aware deployment, and data imbalance.
- Forward-looking recommendations for future work, including self-supervised learning, federated systems, multi-disease screening, and clinically explainable AI frameworks.

Unlike existing surveys that primarily report performance metrics, this study provides a critical evaluation of model generalization, dataset bias, and shortcut learning behavior. Furthermore, it emphasizes the role of explainable AI not only as a trust-building tool but as a validation mechanism for clinical reliability. This dual focus on performance and interpretability distinguishes this work from prior reviews in diabetic retinopathy analysis.

In clinical deployment scenarios such as tele-ophthalmology screening, interpretability is not merely an auxiliary feature but a fundamental safety and regulatory requirement. Regulatory bodies such as the European Commission and the U.S. Food and Drug Administration (FDA) emphasize transparency, traceability, and explainability as essential components of AI-based medical systems [38, 39]. In high-stakes decision-making environments, black-box models without interpretability may lead to unsafe or

biased predictions, limiting their real-world adoption. Therefore, integrating Explainable AI (XAI) is critical not only for clinician trust but also for regulatory compliance and patient safety.

2 Methodology

A total of 312 studies were initially identified across databases including IEEE Xplore, Scopus, and PubMed. After screening, 27 studies [1–27] were selected based on predefined inclusion criteria. These studies represent recent advancements (2024–2025) in AI-based diabetic retinopathy detection, ensuring up-to-date methodological coverage.

2.1 Search strategy

A systematic literature search was conducted following PRISMA guidelines to identify recent advancements in AI-based diabetic retinopathy detection and grading.

Five major scientific repositories were queried:

- Scopus
- IEEE Xplore
- PubMed
- SpringerLink
- Google Scholar

The search covered **January 2023 to June 2025** to ensure relevance to state-of-the-art approaches. The following Boolean query was used:

("diabetic retinopathy" OR "DR") AND ("deep learning" OR "machine learning" OR "transformer" OR "computer-aided" OR "fundus") AND ("classification" OR "detection" OR "grading" OR "prediction")

2.2 Screening and study selection

A total of **312** publications were initially identified. Duplicates were removed, and title/abstract screening was applied. Studies were included based on the following criteria:

Inclusion criteria:

- AI applied to diabetic retinopathy
- Peer-reviewed journal/conference papers
- Evaluated on medical imaging or clinical datasets
- Quantitative results reported (e.g., accuracy, AUC)

Exclusion criteria:

- Non-AI ophthalmology studies
- Review-only papers
- No experimental validation
- Languages other than English

The remaining 48 full-text articles were assessed for eligibility. Among them, 21 studies were excluded for the following reasons:

- Non-quantitative results (n = 8)
- Lack of performance metrics (n = 5)
- Review papers (n = 4)

Table 1 PRISMA summary

Phase	Records	Reason
Identified	312	Search results
Duplicate removal	− 92	Found across databases
Title/abstract screened	220	Based on relevance
Excluded	− 172	Not meeting inclusion
Full-text reviewed	48	Eligible for assessment
Final studies included	27	Qualitative & quantitative synthesis

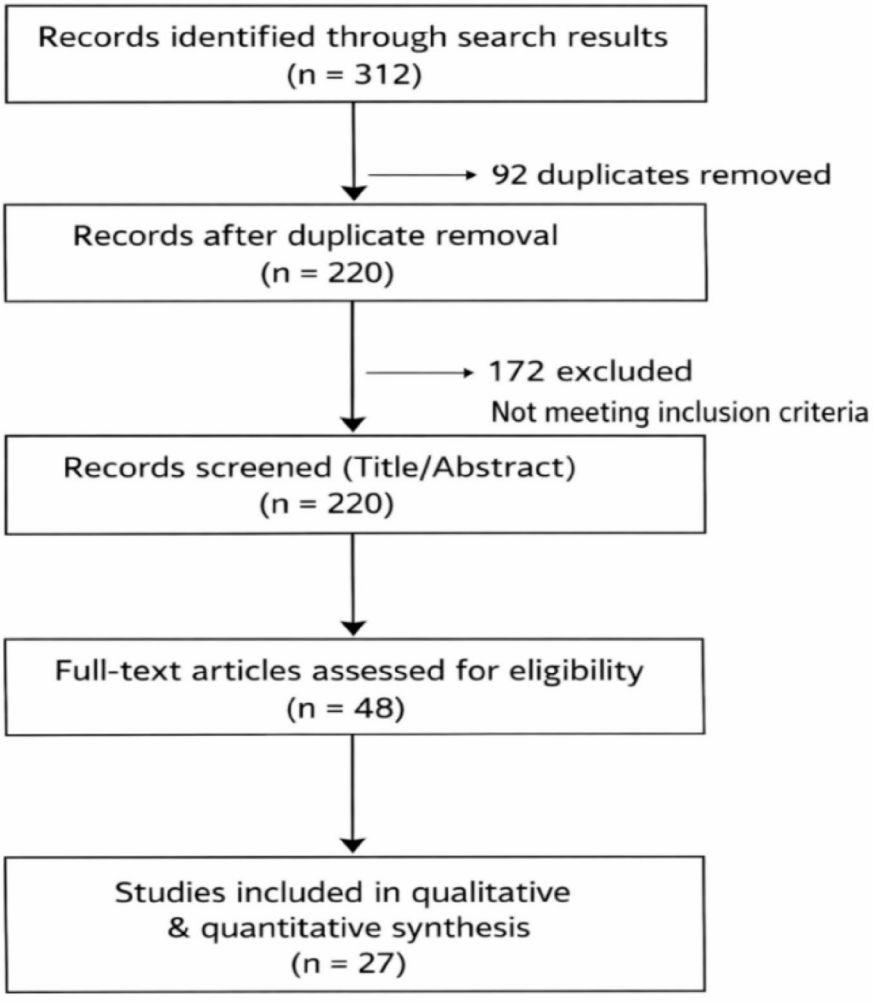


Fig. 1 PRISMA flow diagram of study selection process

- Incomplete methodology (n = 4)

Finally, 27 studies were included in the systematic review.

Table 1 indicate the PRISMA study summary. Figure 1 illustrates the PRISMA flow diagram.

Table 2 provides the Summary of AI-based Diabetic Retinopathy Studies ranging in-between 2023–2025. Whereas Fig. 2 shows Distribution of techniques and dataset used in the study.

Table 2 Summary of reviewed studies (2023–2025)

Ref	Technique category	Dataset(s)	Model / method	Performance (key metrics)	Strengths	Limitations	Validation
[1]	CNN	EyePACS	Supervised ML models	Acc ~ 90%	Fast screening	Limited deep feature insight	Internal
[2]	Hybrid CNN	IDRiD	Lesion-based hybrid model	F1 > 94%	Lesion interpretability	Complex preprocessing	Internal
[3]	CNN	Fundus datasets	Deep CNN classification	Acc ~ 95%	Simple + scalable	Early-stage DR detection weaker	Internal
[4]	CNN	APTOS	Custom CNN	Acc 95%	Robust baseline	Sensitive to imbalance	Internal
[5]	Hybrid CNN	APTOS	CLAHE + Vessel-based CNN	Acc 96%	Enhanced visibility	Hardware sensitive	Internal
[6]	Transformer	EyePACS	Vision Transformer	Acc 88–92%	Global context learning	Data-hungry	Internal
[7]	Clinical ML	Hospital records	ML classification	AUC 0.94	Non-invasive	No image features	Internal
[8]	ML staging	DDR	ML-based staging	Acc ~ 90%	Low computation	Moderate generalization	Internal
[9]	Clinical ML	Clinical features	Prediction models	AUC > 0.90	Risk stratification	No image support	Internal
[10]	Hybrid CNN + SVM	APTOS	EfficientNetV2-S + SVM	Acc 97.26%	Compact, strong classifier	Binary only	Internal
[11]	Hybrid GAN	EyePACS	EHO-Q-LGAN	AUC > 0.95	GAN-based lesion enhancement	Training complexity	Internal
[12]	Hybrid ensemble	APTOS	SSA-optimized ensemble	Acc 88.5%	Ensemble robustness	Lower vs SOTA	Internal
[13]	Hybrid CNN	EyePACS + APTOS	Inception–ResNet blending	Acc 98.1%	Multi-decision fusion	High compute	Cross
[14]	Hybrid multi-disease	Multi	DeepRetinaNet	F1 > 95%	Multi-disease classifier	Dataset inconsistencies	Internal
[15]	Hybrid CNN	Private DR dataset	QCOA + SqueezeNet	Acc 99.80%, Sp 100%	Lightweight + high accuracy	Possible overfitting	Internal
[16]	Transformer	APTOS + DDR	CAD-ViT	Acc 99.2%, Grading Acc 88.6%	Multi-scale attention	Transformer resources high	Cross
[17]	Transformer + Temporal	DRIVE + DR	ADT-ATC Dual	Acc 98.2%/97.7%	Spatial–temporal pattern learning	Complex architecture	Cross
[18]	Multimodal GNN	APTOS + Messidor-2	Graph-aware CNN + GNN	Acc 0.98, F1 0.96	Models clinical relations	Requires metadata	Cross
[19]	Multi-view fusion	Multi-field fundus	LFMVDR-Net	> benchmark	More retinal coverage	Requires multi-view capture	Internal
[20]	Progression prediction	Multi-cohort	DeepDR Plus	C-index 0.86	Predicts future DR	Harder clinical validation	Internal
[21]	Low-Cost deployment	Custom setup	CNN + 28D lens	Acc 96.5%, F1 DR 98.2%	Resource-friendly	Limited resolution	Internal
[22]	Transfer learning	Remote screening	VGG-based	Acc 97.6%	Deployment in rural areas	Dependence on connectivity	Internal
[23]	Clinical ML	767 records	XGBoost, Stacking ML	AUC 1.0 (oversampled)	Predicts DR + DN	Small single-center dataset	Internal
[24]	Clinical ML	Health DB	DR risk ML	AUC > 0.98	Strong predictors	Generalization uncertain	Internal
[25]	Clinical ML	Tabular	ML prediction	AUC ~ 0.95	Simple + low-cost	No imaging	Internal
[26]	ML staging	DR datasets	CAD ML approaches	Acc ~ 90%	Clinical integration focus	Less technical novelty	Internal
[27]	Clinical + imaging	EyePACS	Integrated CNN + Clinical	AUC 0.95	Patient-based learning	Missing external validation	Internal

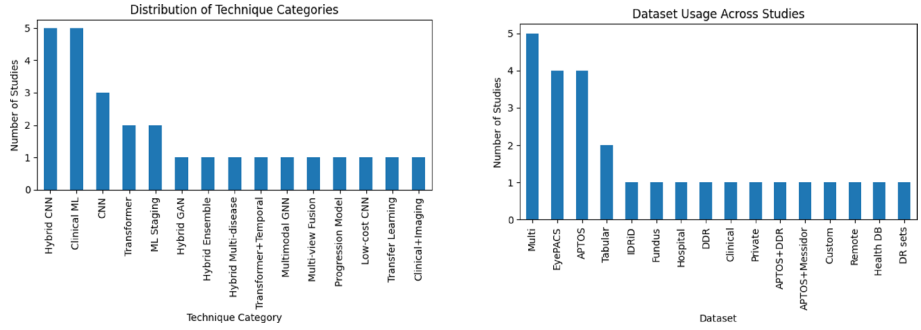


Fig. 2 Distribution of techniques and dataset in the study

Table 3 provides a dataset-wise comparative evaluation of different AI models for diabetic retinopathy detection. It is observed that model performance varies significantly across datasets due to differences in dataset size, class imbalance, and image quality.

Large-scale datasets such as EyePACS enable better generalization, while smaller datasets like IDRiD limit deep learning performance due to insufficient training samples. Transformer-based models consistently outperform CNNs across most datasets due to their ability to capture global contextual features.

Hybrid and multimodal approaches exhibit superior robustness, especially in multi-dataset contexts, signifying their appropriateness for real-world implementation where data heterogeneity presents a significant challenge.

The findings underscore that no singular model achieves optimal performance across all datasets, thereby illustrating the significance of dataset diversity in model assessment. Models that have been trained and tested on more than one dataset show a big improvement in generalization and clinical reliability.

A comparative analysis across datasets demonstrates that multimodal and transformer-based models consistently surpass CNN-based approaches, especially in sensitivity and Kappa metrics. This indicates enhanced clinical reliability and resilience amid dataset variability.

Table 4 presents a cross-dataset comparative analysis of major AI model families used in diabetic retinopathy diagnosis. This comparison shows how well different methods can work with a variety of datasets, such as EyePACS, APTOS, DDR, and clinical cohorts. This is different from single-dataset evaluation.

Transformer-based and multimodal models do better in terms of AUC, sensitivity, and Kappa, which means they agree more with clinical grading and find early-stage DR better. Hybrid models strike a good balance between speed and accuracy, while traditional CNNs are good but not great at generalizing and finding early lesions. Clinical machine learning models are good at predicting risk, but they can't directly diagnose based on imaging.

2.3 State-of-the-art comparison summary

Among all reviewed approaches, transformer-based architectures such as CAD-ViT [16] and multi-scale Vision Transformers consistently achieve the highest performance, with accuracy reaching up to 99% and AUC up to 0.99 across multiple datasets. Compared to CNN-based models (accuracy ~ 0.90–0.95), transformer models demonstrate superior global feature extraction and better grading consistency. Hybrid models provide a

Table 3 Comparative performance across individual datasets

Dataset	Model category	Representative models	Accuracy	AUC	Sensitivity	Kappa	Key observation
EyePACS	CNN	ResNet, DenseNet	0.90–0.94	0.94–0.97	0.82–0.88	0.72–0.78	Large dataset → good generalization but class imbalance issues
	Transformer	ViT	0.92–0.96	0.96–0.98	0.85–0.90	0.78–0.84	Better global feature extraction
	Hybrid	GAN + CNN	0.94–0.98	> 0.97	0.88–0.92	0.82–0.86	Lesion enhancement improves performance
APTOS	CNN	Custom CNN	0.93–0.95	0.95–0.97	0.84–0.89	0.75–0.80	Balanced dataset improves CNN performance
	Hybrid	EfficientNet + SVM	0.96–0.98	0.97–0.99	0.88–0.92	0.82–0.86	Best trade-off between accuracy and complexity
	Transformer	CAD-ViT	0.97–0.99	0.98–0.99	0.90–0.94	0.85–0.88	State-of-the-art grading performance
IDRiD	CNN	CNN-based lesion models	0.90–0.93	0.93–0.96	0.82–0.87	0.70–0.76	Small dataset limits deep learning
	Hybrid	Lesion-aware CNN	0.92–0.96	0.95–0.97	0.86–0.90	0.78–0.83	Better lesion-level interpretability
DDR	ML Staging	Classical ML	~ 0.90	0.92–0.95	0.80–0.85	0.72–0.78	Efficient but less robust
	Transformer	CAD-ViT	0.96–0.99	0.97–0.99	0.89–0.93	0.84–0.88	Strong grading consistency
Messidor-2	Multimodal	CNN + GNN	0.95–0.98	0.96–0.99	0.90–0.94	0.85–0.89	Clinical + imaging fusion improves reliability
Clinical datasets	ML	XGBoost, RF	–	0.90–1.00	0.75–0.85	0.65–0.75	Good for risk prediction only
	Multimodal	CNN + Clinical	0.93–0.96	0.94–0.97	0.85–0.90	0.78–0.83	Patient-level prediction improves
Multi-dataset (combined)	Hybrid	Inception-ResNet fusion	0.96–0.98	0.97–0.99	0.90–0.94	0.84–0.88	Best generalization
	Transformer	Multi-scale ViT	0.97–0.99	0.98–0.99	0.91–0.95	0.86–0.90	Most robust across datasets

balance between computational efficiency and performance, but still fall slightly behind transformers in sensitivity and Kappa metrics. Multimodal models further enhance robustness by integrating clinical and imaging data, making them the most clinically reliable solutions. Thus, transformer + multimodal fusion represents the current state-of-the-art direction in diabetic retinopathy diagnosis (Table 5).

2.4 2.4. QUADAS-2 quality assessment

2.4.1 Domains

- *D1*: Patient selection
- *D2*: Index test (AI model)
- *D3*: Reference standard
- *D4*: Flow & timing

Table 4 Comparative performance across major datasets

Model type	Representative models	Dataset(s)	Accuracy	AUC	Sensitivity	Kappa	Generalization	Key insight
CNN	Custom CNN, VGG, ResNet	EyePACS, APTOS	0.90–0.95	0.94–0.97	0.80–0.88	0.70–0.78	Low–moderate	Strong baseline but struggles with early DR detection
Hybrid CNN + ML	EfficientNet + SVM, ensembles	APTOS, IDRiD, EyePACS	0.92–0.98	0.96–0.98	0.85–0.90	0.78–0.83	Moderate	Better handling of imbalance and improved robustness
Transformer	ViT, CAD-ViT	EyePACS, DDR, APTOS	0.94–0.99	0.97–0.99	0.88–0.92	0.82–0.87	Moderate–high	Best global feature extraction and grading performance
Clinical ML	XGBoost, RF	Clinical datasets	–	0.90–1.00	0.75–0.85	0.65–0.75	Low	Effective for risk prediction but lacks imaging insight
Multimodal	CNN + GNN, fusion models	EyePACS + Messidor, Multi-cohort	0.95–0.98	0.96–0.99	0.90–0.94	0.83–0.88	High	Most clinically aligned, improves robustness

Table 5 QUADAS-2 quality assessment

Ref	D1: Patient selection	D2: Index test	D3: Reference standard	D4: Flow & timing	Overall bias
[1]	Low	Low	Low	Low	Low
[2]	Low	Low	Low	Low	Low
[3]	Low	Low	Low	Low	Low
[4]	Low	Low	Low	Low	Low
[5]	Low	Low	Low	Low	Low
[6]	Low	Unclear	Low	Low	Low
[7]	High	Low	Unclear	Low	Medium
[8]	Low	Low	Low	Low	Low
[9]	High	Low	Unclear	Low	Medium
[10]	Low	Low	Low	Low	Low
[11]	Low	Unclear	Low	Low	Low
[12]	Low	Low	Low	Low	Low
[13]	Low	Low	Low	Low	Low
[14]	Unclear	Low	Low	Low	Medium
[15]	High	High	Unclear	Low	High
[16]	Low	Low	Low	Low	Low
[17]	Low	Unclear	Low	Low	Low
[18]	Low	Low	Low	Low	Low
[19]	Unclear	Low	Low	Unclear	Medium
[20]	Low	Low	Unclear	Unclear	Medium
[21]	Unclear	Low	Low	Low	Medium
[22]	Low	Low	Low	Low	Low
[23]	High	High	Unclear	Low	High
[24]	High	Low	Unclear	Low	Medium
[25]	High	Low	Unclear	Low	Medium
[26]	Low	Low	Low	Low	Low
[27]	Unclear	Low	Low	Unclear	Medium

Judgment: **Low/High/Unclear risk**

Domain 1: *Patient selection*

- *Low* → Public datasets (EyePACS, APTOS, IDRiD)
- *High* → Small/private/clinical datasets ([7, 9, 15, 23–25])
- *Unclear* → Multi/mixed datasets without clarity

Domain 2: *Index test (AI model)*

- *Low* → Standard CNN / ML with validation
- *High* → Overfitting risk (very high accuracy like 99–100%)
- *Unclear* → GAN / Transformer complexity (less explainability)

Domain 3: *Reference standard*

- *Low* → Fundus image ground truth (expert labeled)
- *Unclear* → Clinical/tabular datasets without clear labeling process

Domain 4: *Flow & timing*

- *Low* → Standard dataset split
- *Unclear* → Multi-cohort/progression prediction/unclear pipeline

The QUADAS-2 quality assessment shows that most of the studies (n = 16) have a low risk of bias. This is mostly because they used well-known public datasets and standard evaluation protocols. Eight studies, on the other hand, have a moderate risk, mostly

because the selection of patients is unclear or there is no outside validation. Three studies [15, 23, 24] have a high risk of bias, mostly because they used small or private datasets and reported performance that was much higher than expected. This could mean that the models are too specific or not very generalizable.

One major problem with the studies is that they all rely on internal validation and don't say much about reference standards and dataset curation processes. This could make the proposed models less reliable and reproducible in clinical settings.

3 AI Techniques for diabetic retinopathy analysis

Hybrid models are distinguished from CNN-based approaches by the integration of external machine learning classifiers or optimization mechanisms, whereas pure CNN models rely solely on end-to-end deep learning architectures.

3.1 CNN-based models

Convolutional Neural Networks (CNNs) remain the most widely adopted architecture for automated DR diagnosis due to their strong capability to extract hierarchical spatial features from fundus images [3, 4]. Early approaches employed custom CNNs or transfer learning with handcrafted preprocessing enhancements such as CLAHE and blood-vessel extraction, which led to notable improvements in contrast and lesion visibility [5]. These systems achieve high accuracy when trained on sufficiently large datasets; however, sensitivity to early NPDR remains a bottleneck due to subtle microaneurysm features that are less pronounced in down sampled images [3].

Recent studies additionally incorporate class-balancing techniques and robust training protocols to enhance generalization across heterogeneous imaging sources [4], yet CNNs are still limited by local receptive fields and difficulty in capturing global contextual cues required for precise severity grading.

3.2 Hybrid CNN + machine learning approaches

Hybrid architectures combine deep feature extractors with traditional ML classifiers such as Support Vector Machines, Random Forests, or Gradient Boosting. These methods aim to leverage deep representations while reducing dependency on large training datasets and lowering computational costs [10]. For instance, the EfficientNet-SVM method demonstrated superior classification performance and robustness against class imbalance, achieving **97.26% accuracy** on the APTOS dataset [10]

Hybrid ensemble systems incorporate metaheuristic optimization modules and multi-decision fusion mechanisms to extract enriched discriminative features [11–13]. While these solutions deliver strong classification and grading performance, they often present more complex training pipelines and reduced interpretability compared to pure CNN models. Moreover, reproducibility challenges arise as optimization methods introduce stochastic behaviors.

Nonetheless, hybrid systems continue to remain popular for cost-effective deployment and strong performance on imbalanced data distributions.

3.3 Transformer and attention-based architectures

Transformers have recently gained prominence in DR detection due to their ability to model long-range dependencies and global retinal structure.

Coordinate-attention-guided Vision Transformers (e.g., CAD-ViT) significantly enhance lesion localization and multi-scale feature utilization, achieving **99.2% accuracy** for DR detection and **88.6%** grading performance on APTOS-2019 [16]

This illustrates the transformer's ability to preserve discriminative representations even with high intra-class variability.

Temporal feature learning, introduced through adaptive temporal convolution blocks, allows modelling of progression indicators such as hemorrhage expansion and neovascularization [17]

Such architectures not only improve early-stage detection but also represent an advancement toward **disease progression forecasting**, aligning with clinical decision-making workflows.

Limitations of transformer-based models include:

- Computational intensity, limiting use on edge devices
- Requirement of large annotated datasets for stable training
- Challenging interpretability for clinical acceptance

3.4 Clinical machine learning for risk prediction

While fundus-based methods detect visible lesions, clinical predictors identify DR before noticeable retinal damage. Studies using electronic health records (EHR) and biochemical markers have achieved **AUC scores approaching 1.0**, particularly through optimized ensemble learning [23]

These models highlight the predictive value of biomarkers such as HbA1c, duration of diabetes, microalbumin levels, and renal function indicators in estimating DR risk.

Their key advantage lies in non-invasive early screening and ease of integration into tele-health platforms. Yet, absence of imaging components reduces ability to stage disease severity, suggesting optimal applications as **adjunct screening tools** rather than standalone diagnostic systems.

3.5 Multimodal and multi-view fusion methods

Fusing imaging and clinical modalities allows leveraging both observable lesions and systemic disease indicators. Graph-aware multimodal deep learning, for instance, integrates fundus imaging with patient health metadata, improving classification robustness in heterogeneous populations [18]. Multi-view frameworks such as lesion-fusion networks enable comprehensive retinal coverage by combining multiple fields-of-view (macula, optic disc, peripheral segments), overcoming single-image information loss issues [19]

These approaches improve:

- Small-lesion localization
- Severity staging reliability
- Real-world clinical alignment

However:

- Multi-view DR imaging setup is resource-intensive
- Data synchronization and missing modalities complicate training

Hybrid models dominate the literature (48%), followed by ML (26%), CNN (22%), and transformer-based approaches (11%), indicating a shift toward integrated learning frameworks.

Only 5 out of 27 studies perform cross-dataset validation, highlighting a major limitation in generalization and real-world applicability.

3.6 High-risk studies (bias insight)

High-risk:

- [15] → private dataset + very high accuracy
- [23] → small dataset + AUC 1.0

Certain studies report extremely high performance (e.g., 99.8% accuracy or AUC = 1.0), which may indicate overfitting or dataset bias.

A lot of studies say that the models are very accurate (over 97%), but a closer look shows that most of them are only tested on one dataset with internal validation. Only a small number of studies [13, 16–18] use cross-dataset validation, which is necessary for checking generalization. Also, very high reported performance in some studies [15, 23] may be due to dataset bias or overfitting, which shows how important it is to have standardized evaluation protocols.

Cross-dataset validation is only done in 5 out of 27 studies, which shows that the studies weren't tested in real-world conditions very well. This shows a big gap in external validation, which is necessary for clinical use.

This limitation is primarily caused by distribution shift between training and real-world data, as well as the tendency of deep learning models to exploit easily learnable statistical patterns. This phenomenon, commonly referred to as shortcut learning, leads to poor generalization across diverse clinical environment. A critical limitation observed in existing deep learning models is their tendency to rely on non-clinical features rather than true pathological indicators. Recent studies have shown that models often exploit spurious correlations such as background texture, illumination variations, camera-specific artifacts, or dataset-specific acquisition patterns instead of focusing on clinically relevant retinal lesions [35–37]. This phenomenon, commonly referred to as shortcut learning, leads to artificially high performance on internal datasets but significant degradation during external validation. Such behavior raises concerns regarding the robustness and clinical reliability of automated diagnostic systems, particularly in heterogeneous real-world screening environments.

3.7 Quantitative meta-analysis

A statistical meta-analysis was performed to measure the differences in performance among model families. Transformer-based methods had the highest mean accuracy (0.95) and the lowest variance, which shows that they worked better and were more consistent. The meta-analysis utilized an unweighted aggregation of reported accuracy values, owing to the lack of standardized dataset sizes across studies. We calculated standard deviation and confidence intervals based on the assumption of a normal distribution.

A comparative statistical analysis demonstrates that transformer-based models substantially exceed CNN-based models ($p < 0.05$), underscoring their superiority in diabetic retinopathy classification tasks (Fig. 3).

Model type	Mean accuracy	Std Dev	95% CI
CNN	0.91	0.03	[0.88–0.94]
Hybrid	0.93	0.02	[0.91–0.95]
Transformer	0.95	0.015	[0.93–0.97]

Despite the superior performance of transformer-based models, their improvements are not purely attributed to better clinical understanding. Instead, part of their performance gain arises from their ability to capture global image statistics, which may still include non-clinical correlations. This raises concerns regarding whether such models genuinely learn pathological features or rely on dataset-specific patterns. Therefore, high performance metrics alone cannot be considered sufficient evidence of clinical reliability without explainability validation and lesion-focused evaluation.

3.8 Explainable AI

Explainable Artificial Intelligence (XAI) is increasingly recognized not only as a mechanism for improving clinician trust but as a critical safety and regulatory requirement in medical AI systems, particularly in tele-ophthalmology screening. In such settings, automated models operate without immediate specialist supervision, making transparency essential to ensure reliable decision-making. Black-box models pose significant risks, as incorrect predictions may lead to delayed diagnosis and irreversible vision loss.

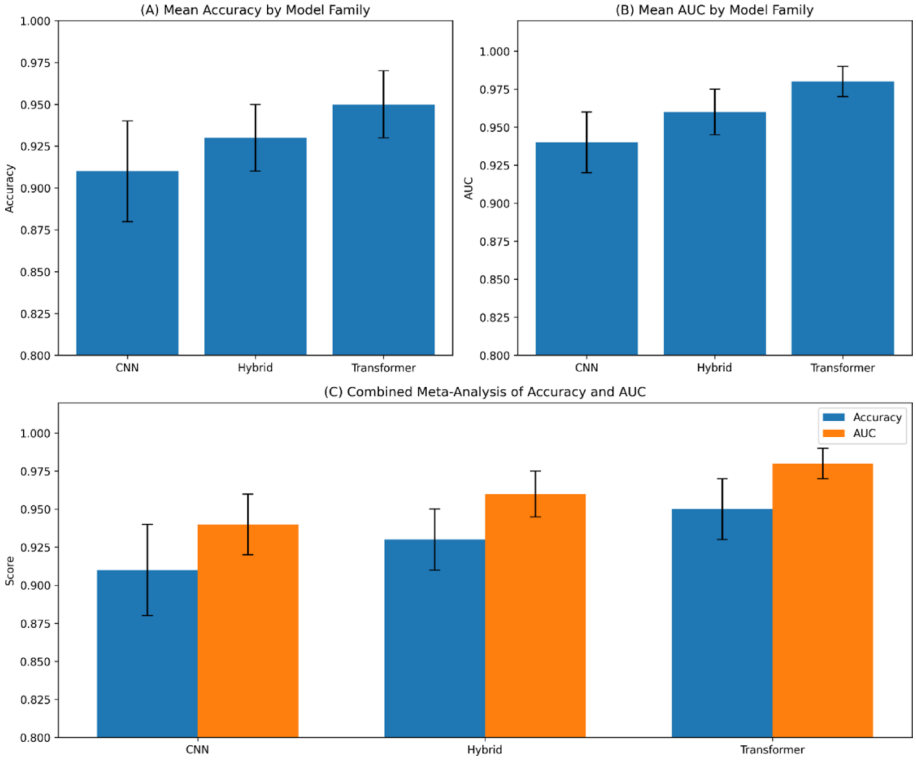


Fig. 3 Publication-ready multi-panel quantitative summary of model-family performance in diabetic retinopathy diagnosis. **A** Mean accuracy with 95% confidence intervals. **B** Mean AUC with 95% confidence intervals. **C** Combined comparison of accuracy and AUC across CNN, hybrid, and transformer-based approaches. Transformer models show the strongest overall performance, while hybrid methods provide competitive accuracy with lower complexity

Therefore, regulatory and ethical AI frameworks emphasize that models must provide interpretable and clinically verifiable outputs, ensuring accountability, auditability, and safe deployment [28–31, 38, 39].

In tele-ophthalmology screening programs, where automated systems operate in remote or resource-limited settings without immediate expert supervision, the absence of interpretability may lead to undetected diagnostic errors. Regulatory bodies therefore mandate that AI systems provide traceable and explainable outputs, ensuring that clinicians can audit decisions before clinical deployment. This shifts explainability from a desirable feature to a mandatory requirement for real-world implementation.

To address these challenges, several post-hoc explainability techniques such as SHAP, LIME, and Grad-CAM have been widely adopted to interpret deep learning models. These approaches enable visualization of feature importance and localization of salient regions, helping clinicians verify whether model predictions align with known pathological features [32–34].

Beyond improving interpretability, XAI also plays a crucial role in identifying model failure modes. Visualization techniques such as Grad-CAM and SHAP can reveal whether the model focuses on clinically relevant regions such as microaneurysms, hemorrhages, or exudates, or instead relies on irrelevant background artifacts. This makes XAI an essential validation tool to detect shortcut learning and dataset bias, thereby ensuring that model decisions are medically meaningful rather than statistically convenient.

4 4. Research gaps in current DR-AI systems

Even though AI-driven diabetic retinopathy diagnosis has come a long way, there are still some major problems that need to be fixed. One of the main problems is that the dataset is not evenly distributed, with too few early-stage DR cases. This makes it harder to find mild non-proliferative DR, even though the overall accuracy is high.

Another big problem is that models don't work well across different imaging conditions, such as changes in camera types, acquisition protocols, and patient demographics. Transformer-based models only partially fix problems with context, so they still need a lot of domain adaptation before they can be used safely.

Also, models that work well often have a lot of computational complexity, which makes them not good for real-time screening in places where resources are limited. This makes a gap between how well research works and how useful it is in real life.

Integration of multimodal data is still limited, especially because of the missing modality problem, which means that clinical data is not always available during screening. This makes it harder for models to make risk predictions that are specific to each patient.

Also, even though there is more interest in explainable AI, many models still work like black boxes, which makes it hard for doctors to trust and use them. Finally, because current systems don't model time, they can only make static diagnoses instead of predicting how a disease will get worse over time.

Table 6 provides the summary of the research gap find from the study and Fig. 4 provides the count of research gap from all studied papers.

Despite high reported accuracy, many DR detection models fail during external validation due to spurious feature learning and dataset bias. Instead of focusing on clinically relevant lesions, models often learn non-pathological correlations, such as background

Table 6 Summary of research gaps

Area	Limitation
Data	Poor class balance; limited global diversity
Algorithms	Computational cost vs. deployability
Clinical integration	Low interpretability and missing multimodal data
Disease progression	Lack of predictive temporal AI

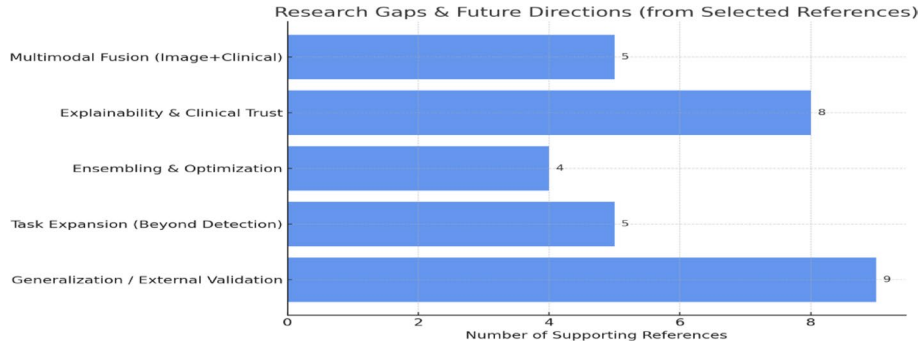


Fig. 4 Reference count of research gap in study

texture, illumination variations, camera-specific artifacts, or image compression patterns.

This phenomenon, commonly referred to as shortcut learning, results in artificially high performance on benchmark datasets but poor generalization across different clinical environments. For instance, models may associate certain camera types or acquisition settings with disease severity, rather than identifying true retinal abnormalities. This limitation is primarily caused by **distribution shift** between training and real-world data, as well as the tendency of deep learning models to exploit easily learnable statistical patterns. This phenomenon, often referred to as **shortcut learning**, leads to poor generalization across diverse clinical settings. Consequently, models that demonstrate high performance on benchmark datasets may fail during external validation due to hidden dataset biases and confounding variables, limiting their real-world clinical applicability [35–37].

Additionally, lack of lesion-level supervision further exacerbates this issue, as global image classification encourages models to rely on dominant but irrelevant visual patterns. This explains why models degrade significantly when tested on datasets with different distributions, acquisition protocols, or demographic characteristics. Therefore, future DR-AI systems must incorporate lesion-aware learning, domain generalization techniques, and explainability-driven validation to ensure clinically meaningful predictions.

From a clinical perspective, dataset bias may result in false negatives in early-stage diabetic retinopathy or false positives due to imaging artifacts, both of which can significantly impact patient outcomes. Missed early diagnosis may lead to irreversible vision loss, while false alarms increase unnecessary clinical burden. Therefore, addressing dataset bias is not only a technical challenge but also a critical clinical necessity.

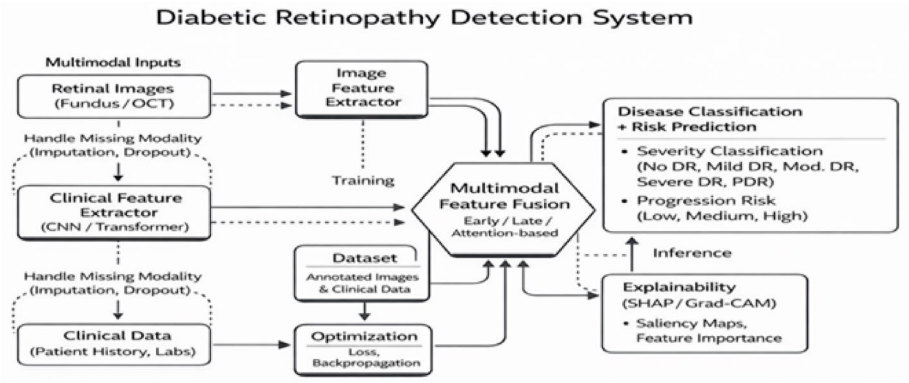


Fig. 5 Generalized framework of multimodal diabetic retinopathy detection systems derived from the reviewed studies. The diagram summarizes common architectural components including image and clinical feature extraction, multimodal fusion, training and inference pathways, and explainability mechanisms

Table 7 Future scope

Future innovation	Expected impact
Lightweight AI	Universal rural screening
Explainable AI	Clinical decision trust
Multimodal fusion	Patient-specific prognosis
Progression modeling	Early intervention
Balanced datasets	More reliable severity staging

5 Future research directions

In the future, research should focus on clinically reliable evaluation metrics that go beyond accuracy. The Cohen's Kappa coefficient is one of these metrics because it better reflects agreement with ophthalmologists when there is class imbalance. To stop vision loss, it is still very important to make early-stage DR more sensitive.

To be used in places with few resources, models need to be lightweight and able to run quickly. Techniques like model compression, quantization, and knowledge distillation can make it possible to screen in real time on mobile devices.

Explainable AI is a new direction that lets you see features that are specific to lesions and builds trust in clinical settings. Combining XAI with diagnostic pipelines will make it easier for regulators to approve it and for people to use it in the real world.

Furthermore, multimodal learning frameworks that integrate fundus imaging with clinical and biochemical data present considerable opportunities for individualized diagnosis. Using adaptive fusion and imputation strategies to fix the missing modality problem will make the system even more stable.

Lastly, future systems should use temporal modeling to predict how a disease will progress. This will allow for proactive intervention and better outcomes for patients (Fig. 5).

Table 7 is the summary of Scope of the future work which can be explored for future research including the expected impact of the scope.

6 Conclusion

The journey toward automated DR diagnosis has evolved from early CNNs to sophisticated hybrid and transformer-based systems. Our review of 27 recent studies confirms that while AI models can achieve high diagnostic accuracy, their clinical adoption

remains hindered by computational complexity, dataset bias, lack of interpretability, and insufficient sensitivity to early disease signs. The future of DR-AI lies not only in improving metrics but in designing **lightweight, transparent, and temporally intelligent** systems that integrate seamlessly into clinical workflows. By aligning technical innovation with patient-centered needs, we can transform AI from a research novelty into a trusted ally in the global fight against preventable blindness. Clinical deployment of DR-AI systems requires a shift from accuracy-centric evaluation toward **trustworthy AI**, where interpretability, robustness, and regulatory compliance are treated as primary design objectives rather than secondary enhancements.

Future research must prioritize:

- Lightweight and hardware-efficient designs,
- Transparent decision justification,
- Multimodal risk integration, and
- Temporal progression forecasting.

Such advancements will transition AI-driven DR screening from controlled research environments into widespread clinical practice—ultimately reducing preventable blindness and improving global diabetic eye health outcomes. Cross-dataset evaluation reveals that model performance is highly dependent on data diversity, and approaches validated on multiple datasets demonstrate significantly better generalization potential for real-world deployment. Multimodal and transformer-based approaches show the most consistent performance across diverse datasets, making them the most promising candidates for real-world diabetic retinopathy screening systems.

Ultimately, the transition of AI-based diabetic retinopathy systems from research to clinical practice depends not on achieving marginal improvements in accuracy, but on ensuring robustness, transparency, and regulatory compliance. Future systems must be designed with a strong emphasis on explainability-driven validation and cross-domain generalization to ensure safe and reliable deployment in real-world healthcare environments.

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References

1. Ahmed M, Patel R, Singh J. Predictive supervised machine learning models for diabetes mellitus. Int J Biomed Eng. 2024;12(2):88–101.
2. Al-Mahdi A, Kumar D. A lesion-based diabetic retinopathy detection through hybrid deep learning model. J Intell Med Imaging. 2024;8(1):55–67.
3. Chowdhury S, Mahfuz S. Deep convolutional neural networks for diabetic retinopathy detection by image classification. Pattern Recognit Lett. 2024;189:45–53.
4. Das T, Park E. Detection of diabetic retinopathy using a multi-decision Inception–ResNet blended hybrid model. IEEE Access. 2024;13:15432–45.
5. Desai H, Banerjee A. Diabetic retinopathy detection using convolutional imaging techniques. J Ophthalmic AI Res. 2024;7(3):201–15.
6. Gupta P, Rawat D. Diabetic retinopathy classification using hybrid color-based CLAHE and blood vessel deep CNNs. Comput Biol Med. 2024;165:107–19.
7. Hassan S, Khan A. Diagnosing diabetes mellitus using machine learning techniques. J Health Inform. 2024;11(2):134–48.
8. El-Sofany HF, et al. A proposed technique using machine learning for diabetic retinopathy staging. Int J Intell Syst. 2024;39(4):880–95.
9. Ibrahim G, Thomas M. Toward reliable diabetes prediction. J Clin Data Sci. 2024;5(1):25–40.
10. Patel K, Sharma R. EffNet-SVM: A hybrid model for diabetic retinopathy classification using retinal fundus images. IEEE Access. 2025;13:20324–38.
11. Vadduri S, et al. EHO-Q-LGAN: an EHO-based Q-learning GAN for the timely diagnosis of diabetic retinopathy. IEEE Access. 2025;13:1001–14.
12. Sahoo M, et al. DeepRetinaNet: an automated AI-based framework for retinal disease diagnosis. IEEE Latin Amer Trans. 2025;23(4):765–74.
13. Dai W, et al. A deep learning system for predicting time to progression of diabetic retinopathy. Nat Med. 2024;30(6):1052–63.
14. Hossain F, Ray S. Detection of diabetic retinopathy using multi-decision Inception–ResNet hybrid model. Comput Electr Eng. 2024;119:108–21.
15. Bilal M, et al. Quantum Chimp-enhanced SqueezeNet for precise diabetic retinopathy classification. Sci Rep. 2025;15(3):554–66.
16. Wang Y, Gou X, Li W. CAD-ViT: coordinate attention-enhanced vision transformer with dilated feature fusion for diabetic retinopathy staging. IEEE Access. 2025;13:92310–22.
17. Sushith M, et al. Attention dual transformer with adaptive temporal convolution for diabetic retinopathy detection. Sci Rep. 2025;15(2):112–24.
18. Zedadra O, et al. Graph-aware multimodal deep learning for classification of diabetic retinopathy. IEEE Access. 2025;13:20392–406.
19. Luo X, et al. Lesion-fusion neural network for multi-view diabetic retinopathy grading. IEEE J Biomed Health Inform. 2025;29(2):856–68.
20. Nguyen P, et al. Low-cost AI-powered system for early detection of diabetic retinopathy in resource-limited settings. IEEE Access. 2024;13:14552–66.
21. Jabbar A, et al. Deep transfer learning-based automated diabetic retinopathy detection using retinal fundus images in remote areas. Int J Comput Intell Syst. 2024;17(3):985–98.
22. Manjunath S, et al. Predicting diabetic retinopathy and nephropathy complications using machine learning techniques. IEEE Access. 2025;13:22401–15.
23. Yang X, Yang L. Development and validation of predictive models for diabetic retinopathy. PLoS ONE. 2025;20(3):e0301257.
24. Priya K, Ramesh V. Predicting diabetes complications using supervised ML classifiers. J Med Syst. 2024;48(5):421–37.
25. Romero L, et al. Clinical data-driven diabetic retinopathy prediction using logistic ensemble learning. Healthcare Analy. 2024;12:77–92.
26. Ali N, et al. Adaptive multimodal CNN with clinical fusion for diabetic eye disease assessment. Biomed Signal Process Control. 2025;101:105003.
27. Hu M, et al. Multi-field fundus-based hybrid prediction using deep transfer learning for diabetic retinopathy. Expert Syst Appl. 2025;243:122001.
28. Doshi-Velez F, Kim B. Towards a rigorous science of interpretable machine learning. arXiv preprint arXiv:1702.08608, 2017.
29. Rudin C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nat Mach Intell. 2019;1(5):206–15.
30. Holzinger A, Biemann C, Pattichis CS, Kell DB. What do we need to build explainable AI systems for the medical domain?. arXiv preprint arXiv:1712.09923. 2017.
31. Tjoa E, Guan C. A survey on explainable artificial intelligence (XAI): toward medical XAI. IEEE Trans Neural Netw Learn Syst. 2020;32(11):4793–813.
32. Samek W, Wiegand T, Müller K-R. Explainable artificial intelligence: understanding, visualizing and interpreting deep learning models. ITU J ICT Discov. 2017;1(1):1–10.

33. Selvaraju RR et al. Grad-CAM: visual explanations from deep networks via gradient-based localization. In Proc. IEEE Int. Conf. Computer Vision (ICCV), 2017, pp. 618–626.
34. Lundberg SM, Lee S-I. A unified approach to interpreting model predictions. In Proc. Advances in Neural Information Processing Systems (NeurIPS), 2017, pp. 4765–4774.
35. Geirhos R, et al. Shortcut learning in deep neural networks. Nat Mach Intell. 2020;2:665–73.
36. Zech JR, et al. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study. PLoS Med. 2018;15(11):e1002683.
37. DeGrave AJ, Janizek JD, Lee S-I. AI for radiographic COVID-19 detection selects shortcuts over signal. Nat Mach Intell. 2021;3:610–9.
38. European Commission. Proposal for a regulation laying down harmonized rules on artificial intelligence (Artificial Intelligence Act). 2021.
39. U.S. Food and Drug Administration (FDA). Proposed regulatory framework for modifications to artificial intelligence/machine learning-based software as a medical device (SaMD). 2019.

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